

First, navigate to `/etc/snort`. Then, create a rule file with the extension `.rules`, and add your rule to the file with all the features you need to test for the attack you want to check for. Save the file. As an example, here is a rule to check a TCP SYN attack (Figure 3), which is named `tcpsyn-task.rules`. The flag is set to S as the intention is to detect SYN packets from an outside network to any port on your home network. If it does detect a packet, it gives a message in the `Alerts` file that the system is 'under SYN flood attack'.

Now you have to include this rule in the Snort configuration file.

So, in the same location of your rules file, find a file named `snort.conf`. Open it using any editor, add the following line, and save it.

```
include /etc/snort/tcpsyn-task.rules
```

Once you are done with this, your rule has been added to Snort. You then run the `snort` command along with your newly added rule. For this, you run the following command:

```
sudo snort -v -c /etc/snort/Snort-test.conf
```

Check your alert message in the log information

Just run the following command to check the output:

```
tail -f /var/log/snort/alert
```

If the alert message that you have given comes up, it means the attack has taken place; otherwise, it hasn't. To show the output, I have generated the attack from another machine of mine. From this output we can infer that a TCP SYN attack has taken place.

Do give it a try!

If you want to learn more about the security of your system, do check out the official documentation of Snort.

```
Alert tcp any any -> any any (flags: S; msg: "under SYN Flood Attack"; sid:100002; rev:1);
```

Figure 3: The rule written for a TCP SYN attack

```
GNU nano 2.9.3
include /etc/snort/tcpsyn-task.rules
```

Figure 4: Include your rule in the configuration file

```
TCP TTL:45 TOS:0x0 ID:65508 Iplen:20 DgmLen:104 DF
***A*** Seq: 0xEB83632E Ack: 0x50915E71 Win: 0x106 TcpLen: 32
TCP Options (3) -> NOP NOP TS: 3567624726 1386502400
=====
WARNING: No preprocessors configured for policy 0.
10/17-08:09:18.818729 186.225.80.194:59614 -> 10.148.0.4:22
TCP TTL:45 TOS:0x0 ID:65509 Iplen:20 DgmLen:52 DF
***A*** Seq: 0xEB836362 Ack: 0x50915E71 Win: 0x106 TcpLen: 32
TCP Options (3) -> NOP NOP TS: 3567624726 1386502400
=====
10/17-08:09:18.820243 10.148.0.4:22 -> 186.225.80.194:59614
TCP TTL:64 TOS:0x0 ID:41042 Iplen:20 DgmLen:52 DF
***A*** Seq: 0x50915E71 Ack: 0xEB836363 Win: 0x1F7 TcpLen: 32
TCP Options (3) -> NOP NOP TS: 1386502461 3567624726
=====
WARNING: No preprocessors configured for policy 0.
10/17-08:09:19.179130 186.225.80.194:59614 -> 10.148.0.4:22
TCP TTL:45 TOS:0x0 ID:65510 Iplen:20 DgmLen:52 DF
***A*** Seq: 0xEB836363 Ack: 0x50915E72 Win: 0x106 TcpLen: 32
TCP Options (3) -> NOP NOP TS: 3567624816 1386502761
=====
```

Figure 5: First output in the terminal

```
TCP TTL:64 TOS:0x0 ID:40195 Iplen:20 DgmLen:40
*****S Seq: 0x4513B4D9 Ack: 0x6C31D03E Win: 0x200 TcpLen: 20
[**] [!100002:1] under SYN Flood Attack [**]
[Priority: 0]
10/17-08:10:58.543536 10.148.0.5:3020 -> 10.148.0.4:80
TCP TTL:64 TOS:0x0 ID:14327 Iplen:20 DgmLen:40
*****S Seq: 0x4E916A3F Ack: 0x4B8F5069 Win: 0x200 TcpLen: 20
[**] [!100002:1] under SYN Flood Attack [**]
[Priority: 0]
10/17-08:10:59.543782 10.148.0.5:3021 -> 10.148.0.4:80
TCP TTL:64 TOS:0x0 ID:28859 Iplen:20 DgmLen:40
*****S Seq: 0xAAD1174 Ack: 0x2070501C Win: 0x200 TcpLen: 20
[**] [!100002:1] under SYN Flood Attack [**]
[Priority: 0]
10/17-08:11:00.543896 10.148.0.5:3022 -> 10.148.0.4:80
TCP TTL:64 TOS:0x0 ID:37290 Iplen:20 DgmLen:40
*****S Seq: 0xBAEABAC Ack: 0x7833054D Win: 0x200 TcpLen: 20
[**] [!100002:1] under SYN Flood Attack [**]
[Priority: 0]
10/17-08:11:01.544078 10.148.0.5:3023 -> 10.148.0.4:80
TCP TTL:64 TOS:0x0 ID:9160 Iplen:20 DgmLen:40
*****S Seq: 0x4B81253E Ack: 0x1F236710 Win: 0x200 TcpLen: 20
```

Figure 6: Alert file thus generated

Learning about the security of your system can be a lot of fun. The deeper you go, the more interesting it gets. The main aim of this article is to give you an idea of how easy and how much fun cyber security can be. 

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